# Loss Functions and Evaluation Metrics for Gesture Classification

Course project presentation for a controller-free fighting game. This deck focuses on training choices, evaluation criteria, and a practical setup for a multi-class ANN gesture classifier.

ANN | MULTI-CLASS CLASSIFICATION | REAL-TIME GAMEPLAY

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# Project Context and Task Goal

The model must classify hand and body gestures into discrete moves such as punch, kick, block, and jump. Inputs come from sensor-derived feature vectors or short time-window sequences, and the output is one label per gesture.

**Core constraints:** low latency for gameplay, robustness to class imbalance, and tolerance for noisy labels.

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# Why the loss choice matters

The loss function shapes what the model learns first. In gesture recognition, that means deciding whether to optimize for stable convergence, rare-move sensitivity, or balanced performance across all classes.

### Balanced data

Sparse Categorical Cross Entropy is usually the clean baseline.

### Imbalanced data

Focal Loss can reduce the dominance of frequent gestures.

### Gameplay impact

False positives and missed gestures both affect responsiveness.

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# Sparse Categorical Cross Entropy

**Definition:** standard cross-entropy computed from integer class labels, so one-hot encoding is not required. It is simple, memory-efficient, and commonly used as the baseline for multi-class classification.

**Strengths:** stable gradients, fast convergence, and easy implementation.

**Limitations:** it can under-emphasize minority gestures when the dataset is imbalanced.

**Mitigation:** class weighting, oversampling, or balanced batch sampling.

**Best use case:** start here when gesture classes are approximately balanced or when you need a reliable first experiment.

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# Focal Loss

Focal Loss modifies cross-entropy to down-weight easy examples and focus learning on hard, misclassified samples. That makes it useful when common gestures dominate the training signal.

**Strengths:** better learning for rare gestures, stronger focus on difficult examples, and less class dominance during training.

**Tradeoffs:** it requires tuning of gamma and optional alpha, and very large gamma values can slow convergence.

**Best use case:** severe class imbalance, or gameplay where minority gestures are critical and cannot be missed.

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# Evaluation metrics: what actually matters

## Accuracy

Overall fraction of correct predictions. Useful for a quick check, but it can hide weak minority-class performance.

## Precision

How often predicted positives are correct. Important when false positives trigger the wrong move.

## Recall

How many true gestures are found. Important when missed inputs make controls feel unresponsive.

**Important note:** accuracy alone can look strong even when the model performs poorly on rare gestures. That is why the metric set must reflect gameplay behavior, not just average correctness.

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# Macro F1 Score as the primary metric

**F1** is the harmonic mean of precision and recall for each class. **Macro F1** averages class F1 scores equally, so every gesture matters the same during reporting.

**Why it is preferred:** it balances false positives and false negatives, and it exposes minority-class performance instead of hiding it behind frequent classes.

**Recommended role:** use Macro F1 for model selection, hyperparameter comparison, and final reporting.

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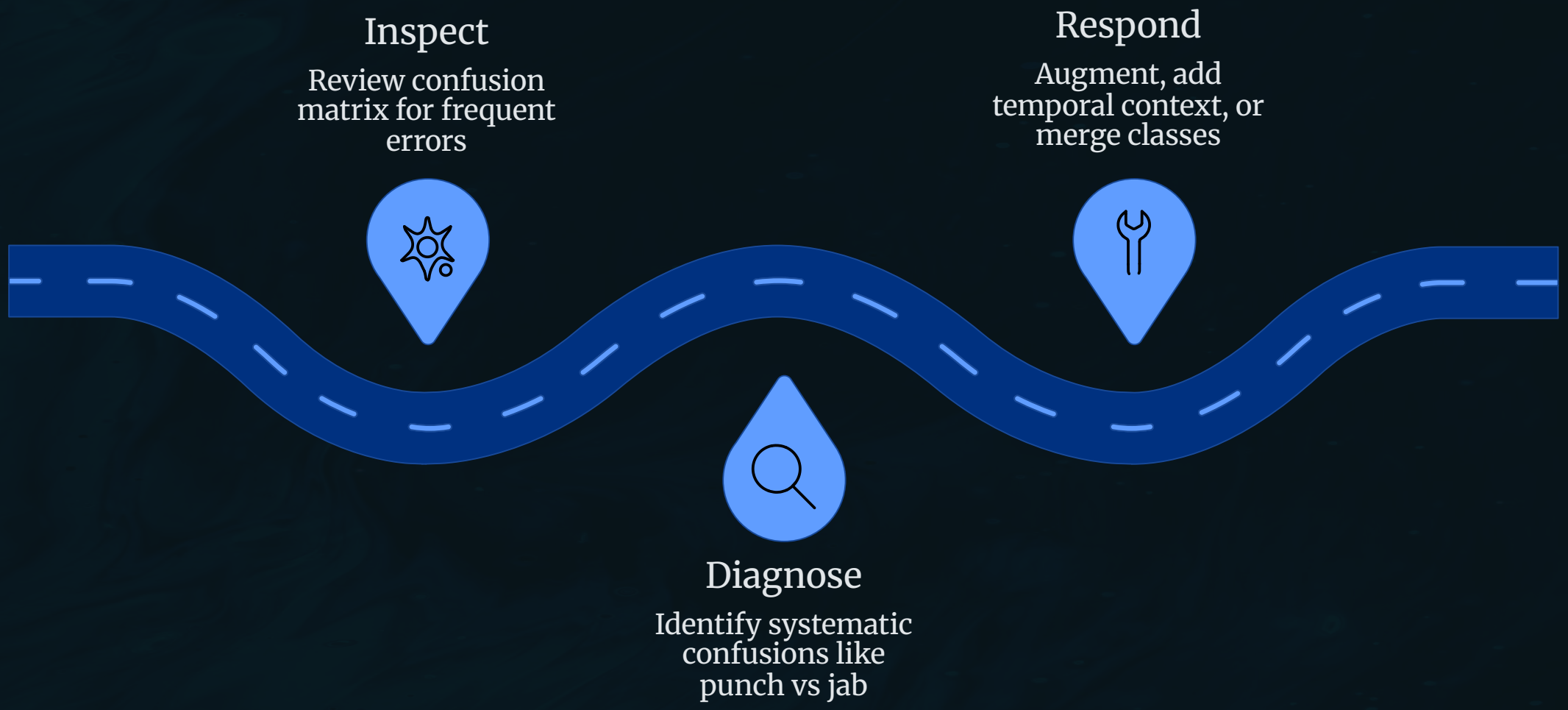
# Confusion matrix: the diagnostic tool




The confusion matrix shows which gestures are being mixed up. In practice, it helps identify recurring errors, guide data augmentation, and reveal whether the model needs more temporal context or more samples for specific moves.

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# Losses and metrics: practical summary

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| **Sparse Cross Entropy** | Stable baseline; best when classes are balanced; pair with class weights if imbalance is mild. |
| **Focal Loss** | Better for severe imbalance and rare gestures; tune gamma and alpha with validation. |
| **Accuracy** | Good for quick checks, but not enough on its own in imbalanced settings. |
| **Macro F1** | Recommended primary metric; treats all classes equally and reveals minority-class behavior. |

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# Recommended setup and next steps

Start with Sparse Categorical Cross Entropy as the baseline. If minority gestures suffer, switch to Focal Loss or combine class weighting with focal loss. Use Macro F1 as the primary metric, report per-class precision and recall, and include a confusion matrix for diagnostics.

## Operational tips

Balanced batch sampling, targeted augmentation for rare gestures, and low-latency inference profiling.

## Why this setup wins

Compared with accuracy-only reporting or unweighted loss alone, it better protects rare moves and gameplay responsiveness.

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